NAM KHANH TRINH

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EDUCATION

University of South Florida — Tampa, FL
B.S. in Computer Science — *Expected May 2030*

Hanoi-Amsterdam High School for the Gifted — Hanoi, Vietnam
High School Diploma; GPA: 9.3 (Gr. 9), 9.5 (Gr. 10), 9.7 (Gr. 11) / 10 — *June 2026*

- **Test Scores:** SAT 1530 (Verbal 770, Math 760), Superscore 1570; IELTS 7.5 (L 8.0, R 9.0, W 7.0, S 6.5)

RESEARCH EXPERIENCE

Undergraduate Research Assistant — Oct 2026 – Present
ERA Lab, University of South Florida (Advisor: Prof. Xiaomin Lin) — *Tampa, FL*

- Research 3D sonar-based object detection from underwater point clouds, building on the lab's SonarVoxNet work
- Study sparse voxel attention mechanisms for efficient feature learning on sparse, noisy 3D sonar data
- Explore multi-modal fusion architectures to improve detection robustness in underwater environments

PUBLICATIONS

Application of Machine Learning Algorithms in Prediction of Hotel Booking Cancellations | *IJSRM* — Sept 2025

- Published in the *International Journal of Scientific Research and Management*, evaluating Random Forest, k-NN, Naive Bayes, SVM, and Decision Tree for cancellation prediction
- Trained and tested on 36,300 customer records (25,000 train / 11,300 test) using 13 features including lead time, average room price, meal plan, and prior cancellations
- Random Forest achieved the highest accuracy at **91.0%**, outperforming k-NN (82.5%), Naive Bayes (82.0%), SVM (80.0%), and Decision Tree (79.0%)

LEADERSHIP & ACTIVITIES

Technical Lead — Oct 2026 – Present
Society of Competitive Programmers (SCP), University of South Florida — *Tampa, FL*

- Design and lead technical workshops on algorithms, data structures, and competitive programming strategy
- Author and review contest problems, test cases, and editorials for correctness and difficulty balance
- Help host and run club coding contests, managing problem sets and contest logistics

PROJECTS

ECO SMART RAIL TRANSIT: AI-Based Railway Safety Monitoring | *Python, Ultralytics YOLO* — Sept 2025

- Built a computer-vision system that detects potential hazards in railway environments for automated safety monitoring
- Owned the full ML pipeline: data preparation, cloud-based YOLO training, and evaluation on held-out images
- **Gold Medal**, INOVA 2025 (49th International Invention Show), Croatia

AI-Activated Robotic Arm | *Python, Ultralytics YOLO* — July 2025

- Trained a YOLO object-detection model and wrote integration code mapping detections to physical robotic-arm movements
- **Gold Medal**, World Invention Creativity Olympic and Conference 2025, Korea

Diabetic Retinopathy Detection | *Python, Computer Vision* — May 2025

- Built an ML pipeline to detect diabetic retinopathy from labeled retinal fundus images, covering preprocessing, training, and evaluation
- **Gold Medal**, Advanced Innovation Global Competition, Singapore

TECHNICAL SKILLS

Languages: Python, C++, JavaScript/TypeScript, SQL, HTML/CSS
ML/AI: PyTorch, TensorFlow, scikit-learn, Ultralytics YOLO, LangChain, NumPy, Pandas, Matplotlib, Seaborn
Research Areas: Computer Vision, Object Detection, 3D Point Clouds, Sonar Perception, Sensor Fusion
Web/Mobile: React, Next.js, Node.js, Express, FastAPI, Flask, React Native, SwiftUI, Tailwind CSS, Vite
Tools & Hardware: Git/GitHub, Arduino, Raspberry Pi, Sensor Modules, Hardware Prototyping
Spoken Languages: Vietnamese (Native), English (Fluent)